

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO5: *Introduction – NumPy Arrays – NumPy Basics- Math – Random – Indexing – Filtering – Statistics – Aggregation – saving data – Data analysis with Pandas – DataFrame – Combining – Indexing – File I/O – Grouping – Filtering – Sorting – Plotting*
(BL 3)

Assessment Tool Weightage: 20%

QUESTIONS:15

Total Marks:25

Annexure A

Pseudocode Writing Task (Algorithm Design)

Code of Conduct:

*I **P.Sai Ganesh Reddy(192525111)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

P.Sai Ganesh Reddy

Annexure A

Short Answer Test

1. Design a pseudocode algorithm to read a dataset from a file and preprocess it using array and DataFrame operations. The algorithm should handle missing values, remove duplicate records, normalize numerical columns, and standardize categorical values. Include steps for indexing, filtering invalid entries, and saving the cleaned dataset. Ensure the algorithm clearly defines input, processing steps, and output while maintaining computational efficiency.
2. Write a detailed pseudocode algorithm to compute statistical measures such as mean, median, standard deviation, and variance from a large dataset. The algorithm should demonstrate how to efficiently use array-based operations, apply filtering to exclude unwanted values, and perform aggregation. Include steps for validating the dataset and handling edge cases such as empty arrays or extreme outliers.
3. Develop a pseudocode algorithm to combine two or more DataFrames containing related data. The algorithm must include steps for aligning indices, merging datasets based on a common key, removing duplicates, and validating the integrity of the merged data. Incorporate grouping and sorting operations, and ensure that constraints such as consistency and completeness of records are maintained.
4. Create a pseudocode algorithm that filters a dataset based on multiple conditions (e.g., value ranges, categories) and then sorts the filtered results. The algorithm should utilize indexing techniques, logical conditions, and efficient sorting methods. Include steps for handling large datasets and optimizing performance while ensuring that the output meets all filtering constraints.
5. Write a pseudocode algorithm for a complete data analysis workflow that includes reading data from files, converting it into DataFrames, performing grouping and aggregation, and generating plots. The algorithm should specify steps for handling file input/output errors, applying transformations, and producing summarized results suitable for visualization. Ensure clarity in each stage from data loading to final output generation.
6. Design a pseudocode algorithm to perform **data cleaning and transformation on time-series data**. The algorithm should handle missing timestamps, interpolate missing values, remove outliers, and resample data at fixed intervals. Include steps for indexing by time, filtering invalid entries, and ensuring chronological consistency while maintaining efficiency.

7. Write a detailed pseudocode algorithm to implement **feature selection** on a dataset. The algorithm should compute correlation between features, remove redundant or irrelevant attributes, and retain significant features. Include steps for normalization, threshold-based filtering, and validation to ensure improved model performance and reduced dimensionality.
8. Develop a pseudocode algorithm to perform **data partitioning for machine learning**, splitting a dataset into training, validation, and testing sets. Ensure randomization, stratification (if required), and balanced class distribution. Include steps for indexing, shuffling, and validating that no data leakage occurs between partitions.
9. Create a pseudocode algorithm to detect and handle **outliers in a dataset** using statistical methods such as Z-score or IQR. The algorithm should identify extreme values, decide whether to remove or cap them, and ensure that the resulting dataset maintains consistency and reliability for further analysis.
10. Write a pseudocode algorithm to implement **data aggregation and summarization** across multiple categories. The algorithm should group data based on specified attributes, compute aggregated metrics (sum, average, count), and generate a summarized report. Include steps for sorting, indexing, and validating the correctness of aggregated results.
11. Design a pseudocode algorithm to perform **data normalization and scaling** on a dataset. The algorithm should support techniques such as Min-Max scaling and Z-score normalization. Include steps for identifying numerical columns, applying transformations, handling missing values, and validating that all features are scaled correctly within specified ranges.
12. Write a pseudocode algorithm to implement **dimensionality reduction using Principal Component Analysis (PCA)**. The algorithm should include steps for standardizing the dataset, computing covariance matrix, extracting eigenvalues and eigenvectors, and selecting principal components. Ensure efficient matrix operations and validation of reduced dimensions.
13. Develop a pseudocode algorithm to perform **data merging and conflict resolution** from multiple sources. The algorithm should detect conflicting attribute values, apply resolution strategies (e.g., priority rules or averaging), and ensure consistency across the final dataset. Include steps for indexing, validation, and handling missing or inconsistent records.
14. Create a pseudocode algorithm to implement **rolling window computations** on a time-series dataset. The algorithm should calculate moving average and rolling variance using efficient indexing and windowing techniques. Include steps for handling boundary conditions, missing values, and optimizing performance for large datasets.

Write a pseudocode algorithm to perform **data validation and integrity checks** on a dataset before analysis. The algorithm should verify data types, detect missing or invalid entries, enforce constraints (such as value ranges), and generate a validation report. Include steps for filtering errors, logging issues, and ensuring the dataset is ready for further processing.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. **Design a pseudocode algorithm to read a dataset from a file and preprocess it using array and DataFrame operations. The algorithm should handle missing values, remove duplicate records, normalize numerical columns, and standardize categorical values. Include steps for indexing, filtering invalid entries, and saving the cleaned dataset. Ensure the algorithm clearly defines input, processing steps, and output while maintaining computational efficiency.**

ANSWER:

INPUT

File: students.csv

Name, Age, Marks, Gender

Ravi, 20, 80, Male

Ravi, 20, 80, Male

Anu, 90, Female

Raj, 21, Male

CODE / PSEUDOCODE

BEGIN

 READ students.csv

 CONVERT data into DataFrame

 HANDLE missing values

 Age → replace with mean

 Marks → replace with mean

 REMOVE duplicate records

 NORMALIZE numerical columns

 STANDARDIZE categorical values

 FILTER invalid records

 SAVE cleaned data

 DISPLAY cleaned dataset

END

OUTPUT

Cleaned Dataset

Name	Age	Marks	Gender
------	-----	-------	--------

Ravi	20	80	Male
------	----	----	------

Anu	20	90	Female
-----	----	----	--------

Raj	21	85	Male
-----	----	----	------

2. **Write a detailed pseudocode algorithm to compute statistical measures such as mean, median, standard deviation, and variance from a large dataset. The algorithm should demonstrate how to efficiently use array-based operations, apply filtering to exclude unwanted values, and perform aggregation. Include steps for validating the dataset and handling edge cases such as empty arrays or extreme outliers.**

ANSWER:

INPUT

Data = [10, 20, 30, 40, 50]

CODE / PSEUDOCODE

BEGIN

 READ data

 IF data is empty THEN

 DISPLAY "Empty Dataset"

 STOP

 END IF

 CALCULATE mean

 SORT data

 CALCULATE median

 CALCULATE variance

 CALCULATE standard deviation

 DISPLAY results

END

OUTPUT

Mean = 30

Median = 30

Variance = 200

Standard Deviation = 14.14

3. **Develop a pseudocode algorithm to combine two or more DataFrames containing related data. The algorithm must include steps for aligning indices, merging datasets based on a common key, removing duplicates, and validating the integrity of the merged data. Incorporate grouping and sorting operations, and ensure that constraints such as consistency and completeness of records are maintained.**

ANSWER:

INPUT

DataFrame 1:

ID Name

1 Ravi

2 Anu

DataFrame 2:

ID Marks

1 80

2 90

CODE / PSEUDOCODE

BEGIN

 READ DataFrame1

 READ DataFrame2

 ALIGN indices

 MERGE using ID

 REMOVE duplicates

 CHECK missing values

 SORT by ID

 VALIDATE merged data

 DISPLAY result

END

OUTPUT

ID Name Marks

1 Ravi 80

2 Anu 90

4. Create a pseudocode algorithm that filters a dataset based on multiple conditions (e.g., value ranges, categories) and then sorts the filtered results. The algorithm should utilize indexing techniques, logical conditions, and efficient sorting methods. Include steps for handling large datasets and optimizing performance while ensuring that the output meets all filtering constraints.

ANSWER:

INPUT

Marks = [45, 80, 65, 90, 35]

Condition: Marks $\geq$ 60

CODE / PSEUDOCODE

BEGIN

 READ dataset

 FILTER values where Marks $\geq$ 60

 SORT filtered values in ascending order

 DISPLAY result

END

OUTPUT

Filtered Marks:

65

80

90

5. Write a pseudocode algorithm for a complete data analysis workflow that includes reading data from files, converting it into DataFrames, performing grouping and aggregation, and generating plots. The algorithm should specify steps for handling file input/output errors, applying transformations, and producing summarized results suitable for visualization. Ensure clarity in each stage from data loading to final output generation.

ANSWER:

INPUT

sales.csv

Product	Sales
A	100
B	200
A	150
B	250

CODE / PSEUDOCODE

BEGIN

 READ sales.csv

 CONVERT data into DataFrame

 HANDLE missing values

 GROUP data by Product

 CALCULATE SUM and AVERAGE

 CREATE summary

 GENERATE plot

 SAVE results

 DISPLAY summary

END

OUTPUT

Product	Total Sales	Average
A	250	125
B	450	225

Plot Generated Successfully

6. **Design a pseudocode algorithm to perform data cleaning and transformation on time-series data. The algorithm should handle missing timestamps, interpolate missing values, remove outliers, and resample data at fixed intervals. Include steps for indexing by time, filtering invalid entries, and ensuring chronological consistency while maintaining efficiency.**

ANSWER:

INPUT

Date	Value
------	-------

01-01-2026	10
------------	----

02-01-2026	12
------------	----

04-01-2026	16
------------	----

05-01-2026	18
------------	----

CODE / PSEUDOCODE

BEGIN

 READ time-series data

 CONVERT Date to DateTime

 SORT by Date

 SET Date as index

 IDENTIFY missing dates

 INTERPOLATE missing values

 DETECT outliers

 RESAMPLE at fixed intervals

 DISPLAY cleaned data

END

OUTPUT

Date	Value
------	-------

01-01-2026	10
------------	----

02-01-2026	12
------------	----

03-01-2026	14
------------	----

04-01-2026	16
------------	----

05-01-2026	
------------	--

7. Write a detailed pseudocode algorithm to implement feature selection on a dataset. The algorithm should compute correlation between features, remove redundant or irrelevant attributes, and retain significant features. Include steps for normalization, threshold-based filtering, and validation to ensure improved model performance and reduced dimensionality.

ANSWER:

INPUT

Features:

Age

Height

Weight

Salary

Correlation Threshold = 0.80

CODE / PSEUDOCODE

BEGIN

 READ dataset

 HANDLE missing values

 NORMALIZE features

 CALCULATE correlation matrix

 FOR each feature pair

 IF correlation > 0.80 THEN

 REMOVE redundant feature

 END IF

 END FOR

 VALIDATE selected features

 DISPLAY selected features

END

OUTPUT

Selected Features:

Age

Weight

Salary

Redundant feature removed: Height

8. **Develop a pseudocode algorithm to perform data partitioning for machine learning, splitting a dataset into training, validation, and testing sets. Ensure randomization, stratification (if required), and balanced class distribution. Include steps for indexing, shuffling, and validating that no data leakage occurs between partitions.**

ANSWER:

INPUT

Dataset = 1000 records

Training = 70%

Validation = 15%

Testing = 15%

CODE / PSEUDOCODE

BEGIN

 READ dataset

 SHUFFLE dataset

 APPLY stratification if required

 SPLIT 70% for training

 SPLIT 15% for validation

 SPLIT 15% for testing

 CHECK for data leakage

 DISPLAY sizes

END

OUTPUT

Training Set = 700 records

Validation Set = 150 records

Testing Set = 150 records

Total = 1000 records

9. Create a pseudocode algorithm to detect and handle outliers in a dataset using statistical methods such as Z-score or IQR. The algorithm should identify extreme values, decide whether to remove or cap them, and ensure that the resulting dataset maintains consistency and reliability for further analysis.

ANSWER:

INPUT

Data = [10, 12, 13, 14, 15, 100]

CODE / PSEUDOCODE

BEGIN

 READ data

 CALCULATE Q1

 CALCULATE Q3

$IQR = Q3 - Q1$

 Lower Limit = $Q1 - 1.5 \times IQR$

 Upper Limit = $Q3 + 1.5 \times IQR$

 IDENTIFY values outside limits

 REMOVE or CAP outliers

 DISPLAY cleaned data

END

OUTPUT

Outlier Detected = 100

Cleaned Data:

10, 12, 13, 14, 15

10. Write a pseudocode algorithm to implement data aggregation and summarization across multiple categories. The algorithm should group data based on specified attributes, compute aggregated metrics (sum, average, count), and generate a summarized report. Include steps for sorting, indexing, and validating the correctness of aggregated results.

ANSWER:

INPUT

Category Sales

A 100

A 200

B 300

B 400

CODE / PSEUDOCODE

BEGIN

 READ dataset

 GROUP data by Category

 CALCULATE SUM

 CALCULATE AVERAGE

 CALCULATE COUNT

 SORT summary

 DISPLAY report

END

OUTPUT

Category Sum Average Count

A 300 150 2

B 700 350 2

11. **Design a pseudocode algorithm to perform data normalization and scaling on a dataset. The algorithm should support techniques such as Min-Max scaling and Z-score normalization. Include steps for identifying numerical columns, applying transformations, handling missing values, and validating that all features are scaled correctly within specified ranges.**

ANSWER:

INPUT

Data = [10, 20, 30, 40, 50]

Method = Min-Max Scaling

CODE / PSEUDOCODE

BEGIN

 READ dataset

 FIND minimum value

 FIND maximum value

 FOR each value

$$\text{NewValue} = (\text{Value} - \text{Minimum}) /$$
$$(\text{Maximum} - \text{Minimum})$$

 END FOR

 VALIDATE scaled values

 DISPLAY result

END

OUTPUT

Original: 10 20 30 40 50

Scaled:

0.00 0.25 0.50 0.75 1.00

12. Write a pseudocode algorithm to implement dimensionality reduction using Principal Component Analysis (PCA). The algorithm should include steps for standardizing the dataset, computing covariance matrix, extracting eigenvalues and eigenvectors, and selecting principal components. Ensure efficient matrix operations and validation of reduced dimensions.

ANSWER:

INPUT

Dataset:

Feature1

Feature2

Feature3

Feature4

Number of Components = 2

CODE / PSEUDOCODE

BEGIN

 READ dataset

 HANDLE missing values

 STANDARDIZE features

 CALCULATE covariance matrix

 CALCULATE eigenvalues

 CALCULATE eigenvectors

 SORT eigenvalues

 SELECT first 2 components

 TRANSFORM dataset

 VALIDATE dimensions

 DISPLAY reduced dataset

END

OUTPUT

Original Features = 4

Reduced Features = 2

Principal Components:

PC1

PC2

13. **Develop a pseudocode algorithm to perform data merging and conflict resolution from multiple sources. The algorithm should detect conflicting attribute values, apply resolution strategies (e.g., priority rules or averaging), and ensure consistency across the final dataset. Include steps for indexing, validation, and handling missing or inconsistent records.**

ANSWER:

INPUT

Dataset 1:

ID Name

1 Ravi

2 Anu

Dataset 2:

ID Name

1 Ravi

2 Ananya

CODE / PSEUDOCODE

BEGIN

 READ both datasets

 IDENTIFY common ID

 MERGE datasets

 DETECT conflicting values

 IF conflict exists THEN

 APPLY priority rule

 END IF

 REMOVE duplicates

 HANDLE missing values

 VALIDATE data

 DISPLAY final dataset

END

OUTPUT

ID Name

1 Ravi

2 Ananya

Conflict detected for ID = 2

Resolution rule applied

14. Create a pseudocode algorithm to implement rolling window computations on a time-series dataset. The algorithm should calculate moving average and rolling variance using efficient indexing and windowing techniques. Include steps for handling boundary conditions, missing values, and optimizing performance for large datasets.

ANSWER:

INPUT

Data = [10, 20, 30, 40, 50]

Window Size = 3

CODE / PSEUDOCODE

BEGIN

 READ time-series data

 SET window size = 3

 FOR each position

 SELECT current 3 values

 CALCULATE moving average

 CALCULATE rolling variance

 END FOR

 HANDLE boundary values

 DISPLAY results

END

OUTPUT

Data	Moving Average
------	----------------

10,20,30	20
----------	----

20,30,40	30
----------	----

30,40,50	40
----------	----

15. Write a pseudocode algorithm to perform data validation and integrity checks on a dataset before analysis. The algorithm should verify data types, detect missing or invalid entries, enforce constraints (such as value ranges), and generate a validation report. Include steps for filtering errors, logging issues, and ensuring the dataset is ready for further processing.

ANSWER:

INPUT

Dataset:

Age Marks

20 80

21 90

-5 70

22 NULL

CODE / PSEUDOCODE

BEGIN

 READ dataset

 CHECK data types

 CHECK missing values

 CHECK duplicate records

 CHECK invalid values

 CHECK Age range

 FILTER invalid records

 LOG errors

 GENERATE validation report

 DISPLAY valid dataset

END

OUTPUT

Invalid Age detected: -5

Missing Marks detected for Age 22

Validation Report:

Valid Records = 2

Invalid Records = 2

Missing Values = 1